Exercises

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Question

What of the following statement holds **False** for word embedding models?

1. FastText encodes subwords.
2. GloVe exploits occurrence-based statistics.
3. GloVe do not support domain adaptation.
4. WordVec relies on a Deep Feed-forward Neural Network.
5. Word2Vec supports domain adaptation.

Answer

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5. Word2Vec supports domain adaptation.

It relies on a single-layer NN

Answer

What of the following statement holds False for word embedding models?

1. FastText encodes subwords. **It is a contribution w.r.t. Word2Vec**
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It is not possible to adapt to a new corpus without re-training

Answer

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The training can continue over a new corpus to specialize the domain (no global statistics)

Question

Considering the Word2Vec model:

1. Describe its inner working and main goals

Answer

Considering the Word2Vec model:

1. Describe its inner working and main goals

- The word embedding model generates one vector representation per word in the vocabulary.
- Word vectors are not contextualized (unlike, e.g., ELMO)
- It relies on self-supervised training using sliding window.
- Architectures: Skip-gram and CBOW.
- The output of the model is a matrix containing the mapping between each word in the vocabulary and its corresponding vector.

Question

Considering the Word2Vec model:

2. Illustrate in details one of the training procedures (at your choosing).

Answer

Considering the Word2Vec model:

2. Illustrate in details one of the training procedures (at your choosing).

6.2: The CBOW training procedure:

- The context is represented as a bag of the words contained in a fixed-size window aligned with the target word.
- The distributed representations of the context is used to predict the word in the middle of the window.
- The classification model exploits the context to predict the correct target word.
- *Sketch of the architecture.*

Question

Considering the Word2Vec model:

3. Discuss (at least) one of the main drawbacks of the model.

Answer

Considering the Word2Vec model:

3. Discuss (at least) one of the main drawbacks of the model.

Limitation: word representations are not contextual.

The cat eat the $\texttt{mouse}^1$

The $\texttt{mouse}^2$ is not working anymore.

$\mathrm{W2V}(\texttt{mouse}^1) = \mathrm{W2V}(\texttt{mouse}^2)$

Question

Considering the Word2Vec model:

4. Enumerate the key differences with FastText.

Answer

Considering the Word2Vec model:

4. Enumerate the key differences with FastText.

FastText creates n-gram embeddings and generates word vectors using compositionality.

E.g. (with n-gram = 4),

$$V(\texttt{processing}) = \mathrm{FT}(\texttt{<pro}) + \mathrm{FT}(\texttt{proc}) + \mathrm{FT}(\texttt{roce}) + \ldots + \mathrm{FT}(\texttt{ing>}) + \mathrm{FT}(\texttt{processing})$$

Question

Considering the InferSent model:

1. Describe the main goal and the task on which it is trained

Answer

Considering the InferSent model:

1. Describe the main goal and the task on which it is trained

- The main goal of InferSent is to create a semantically meaningful sentence representation.
- It is trained on the inferencing task, which involves determining teh truth value (true, false or neutral) of a hypothesis based on a provide premise

Question

Considering the InferSent model:

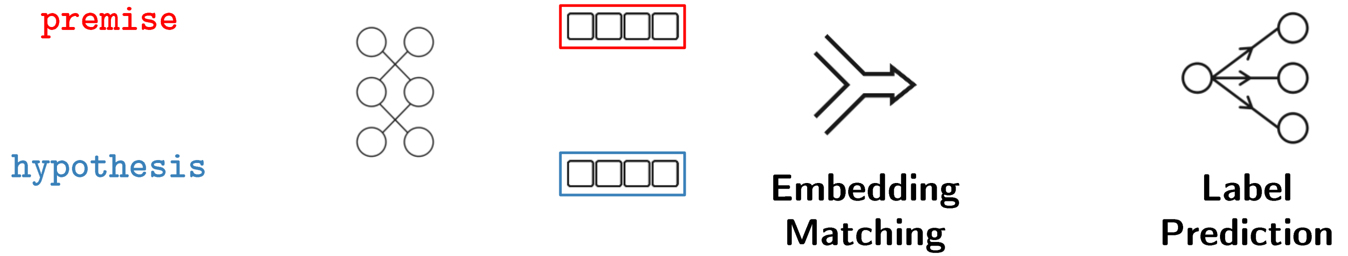
2. Describe the training pipeline (from input data to final model)

Answer

Considering the InferSent model:

2. Describe the training pipeline (from input data to final model)

- A word embedding model (e.g., Word2Vec, GLOVE) is applied to the words of the sentences
- Word embeddings are fed to an embedding model (e.g., LSTM, GRU)
- An embedding matching strategy is applied to predict the correct label among the 3 possible outcomes (entailment, contraddiction, neutral)
- Sketch of the pipeline:



Question

Considering the InferSent model:

3. Indicate at least two embedding matching strategies

Answer

Considering the InferSent model:

3. Indicates at least two embedding matching strategies reporting also the final size with respect the input vectors

- **Concatenation**: the vector representations of the 2 sentences (premise and hypothesys) are concatenated to obtain a unique vector (with doubled size with respect the original ones)
- **Element-wise product**: corresponding elements of the two vectors are multiplied to obtain a new vector of the same size

Question

What of the following statement holds **False** for Doc2Vec-DM?

1. Word and paragraph vector can be combined to predict the next word in a context
2. Paragraph vector always appears in the sliding window
3. All the weights for paragraphs and classification are fixed when you predict a new word vector
4. Every paragraph is mapped into a unique vector

Answer

What of the following statement holds **False** for Doc2Vec PV-DM?

1. Word and paragraph vector can be combined to predict the next word in a context
2. Paragraph vector always appears in the sliding window
3. All the weights for paragraphs and classification are fixed when you predict a new word vector
4. Every paragraph is mapped into a unique vector

It is the opposite!

Answer

What of the following statement holds **False** for Doc2Vec-DM?

1. Word and paragraph vector can be combined to predict the next word in a context

   Yes, they can be averaged or concatenated

2. Paragraph vector always appears in the sliding window
3. All the weights for paragraphs and classification are fixed when you predict a new word vector
4. Every paragraph is mapped into a unique vector

Answer

What of the following statement holds **False** for Doc2Vec-DM?

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Window slides on word vectors while paragraph vector is fixed

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4. Every paragraph is mapped into a unique vector

Unique representation, as for the words

Acknowledgements and copyright license

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        - https://dbdmg.polito.it
        - https://smartdata.polito.it

Thank you!